



Locating Sea Ambulances to Respond to Emergencies of Vulnerable Populations. Case of Cartagena Bay in Colombia

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Abstract. Emergency services are an important element for healthcare assistance because its rapid response to transportation is essential for saving lives. Cartagena de Indias presents some weaknesses in terms of covering the demand for emergency transfers from insular areas due to the non-existence of terrestrial routes to hospitals and healthcare facilities. This study aims to determine the optimal location for sea ambulances using a mixed-integer linear programming model. A three-staged methodology allowed to select optimal locations, reducing response times of emergencies for vulnerable populations considering available healthcare facilities and maritime safety requirements.

Keywords: Location · Sea ambulances · Mixed integer linear programming · Healthcare

1 Introduction

Emergency services are an important element for healthcare assistance because its rapid response in transportation is essential for saving lives [4, 20].

A critical aspect for these services is the physical location of the ambulances. The typical service provided by an ambulance involves: (i) receiving an incoming emergency call and assessing the situation, (ii) dispatching an ambulance to the emergency site, if required, (iii) providing assistance at the site by the emergency medical team, and (iv) transferring the patient to the nearest hospital [1].

Location problems have received substantial attention in the literature because they are very useful in practice, and because of the difficulty in finding optimal solutions [19]. These problems are classified into three major groups: (i) Flat models, in which demand sites are located at discrete points on the plain and the facilities can be located on any point [3]; network models, in which demand is produced in the nodes of a network and the facilities may be located

on the network's nodes [7,13]; and discrete models, in which demand is produced in the nodes of the supply chain, and the facilities can be located based on a set of nodes and the distance between the locations of the facilities and the demand points are represented by a distance matrix [10]. These are the most common models used in the facility location literature.

Discrete location models are sub-divided into four major groups: (i) The median- p models, whose objective is to minimize the total or average distance between the demand nodes and the location of the facility to which they are assigned [14]; (ii) p -center models, which minimize the maximum distance between a demand node and the nearest facility, also known as a minimax location problem; (iii) Maximum coverage models, which are aimed at maximizing the coverage of a facility to cover demand by location [13,15]; and (iv) Fixed cost models, which seek to minimize the cost of opening while ensuring coverage for all demand points [2].

A literature review on healthcare facility location is provided by [1]. Ambulance location problems are generally considered as Maximum Expected Coverage Location Problems (MEXCLP), developed by [11].

Schmid et al. [17] studied the problem of locating ambulance vehicles, allowing covering and responding to emergencies efficiently, depending on the time of travel. To solve the problem, they used the metaheuristic "Variable Neighborhood Search" (VNS).

Zaffar et al. [20] presented a simulation model to assess different policies on coverage, survival and response times as a function of ambulance location.

Dibene et al. [12] presented an application of ambulance location in the city of Tijuana, Mexico, considering the information on possible locations for ambulance bases, demand and priorities, demand scenarios, possible points of demand, and average travel times. This study demonstrated improvements in response and coverage.

Belanger et al. [5] presented a review of optimization models and trends regarding location, relocation and dispatching of medical emergency vehicles.

The literature also includes the problem of helicopter location for the healthcare sector. [8] studies the case of locating helicopters and transfer points to receive patients from land ambulances to then transfer them to hospitals. [16] studies the problem of ambulance vehicle location but considering unique stochastic demands in different time periods and at different points.

Shahriari et al. [18] proposed a dual-objective model for the location of ambulance bases, air bases and heliports subject to uncertainty in transfer times between demand points and hospitals, with penalties based on time windows. The first objective was minimizing the maximum time of transfer under the assumption that the times are stochastic.

Overall, the reviewed literature indicates that the problem of location of ambulance boats has not yet been considered. This topic is only mentioned in some studies about healthcare services in remote islands [9]. This article is intended to be a contribution to the scientific literature, proposing a three-stage methodology that allows the selection of optimal locations for ambulance boats, considering the available healthcare facilities and marine safety requirements.

The aim of this paper is showing the results for the definition of optimal locations for ambulance boats, ensuring coverage and a faster response for vulnerable populations in case of emergencies in the insular area of Cartagena Bay (Colombia) using an MIP optimization model.

This paper is organized as follows: Sect. 2 shows a brief explanation of the implemented methodology, Sect. 3 shows the main results for every stage. Finally, Sect. 4 exposes the main conclusions and opportunities for further research.

2 Methodology

A three-stage methodology allowed to identify and characterize vulnerable populations and healthcare centers with access to the bay and proposing optimal location for ambulance boats using an MIP model:

2.1 Characterization of Healthcare Facilities

This stage involved identifying and characterizing the hospitals that have access by sea to the Cartagena Bay and have the capacity to respond to emergencies by sea. Basic requirements for ambulance boats and regulations established by maritime authorities were considered and the possible docks were identified using the Directorate General Maritime (DIMAR) registry.

2.2 Characterization of Vulnerable Populations

In this stage, the vulnerable populations that can be covered by sea in case of an emergency was identified. Information was gathered through interviews and historical records of assistance provided to the communities by the Coast Guard. In this case, the term “vulnerable” refers to populations with the highest emergency occurrence and accidental rates. Statistics for emergencies and assistance covered by sea were calculated.

2.3 Optimal Location for Ambulance Boats

This stage aimed to determine the location of the sea ambulances of Cartagena Bay, taking into consideration the available healthcare facilities and vulnerable populations identified the first two stages. An Mixed Integer Linear Programming model (MILP) is proposed to define the optimal location for the ambulance boats, considering a from/to travel time matrix.

3 Results

Two hospitals with access to the Cartagena Bay by sea were identified:

- Hospital Nuevo de Bocagrande is a high-complexity healthcare provider that offers emergency services, general hospitalization for adults, intensive and intermediate care units for adults, and other services. This hospital is rated with levels III and IV of assistance complexity.
- Hospital Naval de Cartagena is a Military Healthcare facility, whose declared mission is to provide quick and good services to users when required. In addition to the services offered by other hospitals in the city, Hospital Naval offers multi-stage hyperbaric medicine, specialized treatment of degenerative diseases, traumas, burns, Parkinson's, diving accidents, diabetes complications, encephalopathies, wound healing and ulcers. Hospital Naval de Cartagena is rated as Complexity Level III–IV.

The identified vulnerable populations are shown in Table 1.

Table 1. Insular vulnerable populations

Town	Area	Inhabitants	Risk
Tierra Bomba	Rural district	1460	Middle
Punta arenas	Rural district	352	Middle
Caño de Oro	Rural district	1239	Middle
Bocachica	Rural district	3028	High
Isla de manzanillo	Comuna 10	2180	Middle
Bosque	Comuna 10	12902	Middle
Bocagrande	Comuna 1	14376	Low
Manga	Comuna 1	16005	Low
Contecar-Mamonal	Comuna 11	1900	High
Cotecmar-Mamonal	Comuna 11	1800	High
Centro	Comuna 1	4173	Low
Castillo Grande	Comuna 1	7330	Low

Table 2 shows the historical data on support for evacuations provided by the Coast Guard Unit to these communities, for a total of 186 evacuations assisted between 2013 and 2017. The community that most assistance has received in recent years was Bocachica, with a total of 137 evacuations, followed by the community at Caño de Loro, with 22 evacuations assisted.

The Emergency Regulation Center (CRUE, by its acronym in Spanish) provided data on the activities performed by the Sea Ambulance ‘DADIS’ since 2016, as indicated in Table 3. The sum of events assisted by the ambulance in the Cartagena community in 2016 and 2017 was 1087 transfers by sea.

Overall, in these populations from 2013 to 2017 there were 1273 evacuations by sea, the great majority of which took place in the towns of Bocachica, Caño de Loro and Punta Arenas. It should be noted that the Cartagena Bay area includes the Mamonal industrial park and CONTECAR, which have large numbers of employees, as well as other neighborhoods along the Cartagena Bay. Table 4

Table 2. Evacuation support provided by the coast guard in 2013–2017

Town	2013	2014	2015	2016	2017	Total
Bocachica	62	57	10	6	2	137
Caño de Loro	0	16	5	1	0	22
Punta Arenas	0	8	2	0	0	10
Basurto	0	1	0	0	0	1
Tierra Bomba	0	0	0	10	5	15
Bocagrande	0	1	0	0	0	1
Total	62	83	17	17	7	186

Table 3. CRUE statistics per year for sea assistance in 2016–2017

Town	2016	2017	Total
Bocachica	351	381	732
Bocagrande		1	1
Caño de Loro	152	176	328
Centro	1		1
Punta Arenas	7	18	25
Total	511	576	1087

displays the geographic coordinates of the hospitals with access by sea and the vulnerable populations.

Given the geography of the city itself and the travel times in the city of Cartagena de Indias, combined with heavy vehicle traffic, it is necessary to locate ambulance boats to respond to maritime emergencies in order to improve response times in emergencies. A mixed integer linear programming model was implemented to optimally determine the location of the sea ambulances to cover emergencies of the vulnerable populations and to minimize response and travel times to the main hospitals. This kind of model was selected considering that MIP models are one of the most used in literature for solving ambulance location problems.

The model variables and parameters are listed in Table 5. The objective function seeks to maximize coverage of the vulnerable population according to the risk level of each town (Eq. 1).

$$\text{Max } Z = \sum_{j \in P} w_j * p_j * x_j \quad (1)$$

$$\sum_{i \in L} y_i \leq q \quad (2)$$

$$x_j \leq \sum_{i \in L} y_i * a_{i,j} \forall (j \in P) \quad (3)$$

$$y_i \in \{0, 1\} \forall i \in L; x_j \in \{0, 1\} \forall j \in P \quad (4)$$

Table 4. Geographic coordinates of the hospitals with access by sea and the vulnerable populations

Type	Place	Latitude	Longitude
Hospitals	Nuevo Hospital Bocagrande	10 23' 40.89"N	75 33' 21.14"W
	Hospital Naval	10 25' 02.51"N	75 32' 58.87"W
Vulnerable Populations	Isla Manzanillo	10 23' 34.29"N	75 31' 57.41"W
	Punta Arenas	10 21' 58.04"N	75 33' 13.29"W
	Contecar – Mamonal	10 22' 28.46"N	75 30' 34.14"W
	Coctemar – Mamonal	10 19' 34.79"N	75 30' 36.05"W
	Caño de Loro	10 20' 30.09"N	75 32' 40.78"W
	Bocachica	10 19' 27.01"N	75 34' 40.53"W
	Tierra Bomba	10 22' 39.76"N	75 34' 25.56"W
	Castillo Grande	10 23' 43.62"N	75 33' 02.61"W
	Bosque	10 23' 53.98"N	75 31' 27.27"W
	Manga	10 24' 41.15"N	75 32' 25.12"W
	Centro	10 25' 19.44"N	75 32' 57.16"W
	Bocagrande	10 24' 10.90"N	75 33' 07.83"W

Table 5. Variables and parameters of the proposed model

Symbol	Description
P	Set of indexes identifying vulnerable populations to be covered by sea ambulances
H	Set of hospitals with maritime access
L	Set of indexes identifying all possible locations for sea ambulances. $L = P \cup H$
i	Index for possible locations, where $i \in L$,
j	Index for vulnerable populations locations, where $j \in P$
p_j	Number of inhabitants for population j
w_j	Weight associated to the risk of population j
$t_{i,j}$	Travel time from potential location $i \in L$ to town $j \in P$
$tr_{i,j}$	Travel time from potential location $i \in L$ to town $j \in P$,
	Plus travel time to the nearest hospital from town $j \in P$ (i.e., $t_{i,j} + \min_{k \in P}(t_{j,k})$)
NS	Service level
q	Number of available sea ambulances
$a_{i,j}$	Binary variable that is equals to 1 if $tr_{i,j} \leq NS$, and 0 otherwise
y_i	Binary variable that is equals to 1 if a sea ambulance is located in location $i \in L$, and 0 otherwise
x_i	Binary variable that is equals to 1 if vulnerable population $j \in P$ can be covered by an ambulance in less than NS , and 0 otherwise

The first constraint (Eq. 2) controls the number of available ambulances, this is, no additional ambulances than the existing ones can be added.

The second constraint (Eq. 3) helps to meet the response time defined for covering an emergency in a vulnerable population.

The travel routes were determined for all the nodes, based safety standards established by DIMAR, and considering that the emergency boat seeks to perform the transportation as quickly as possible, depending on its technical characteristics and the zone of operations. These routes are shown in Fig. 1.

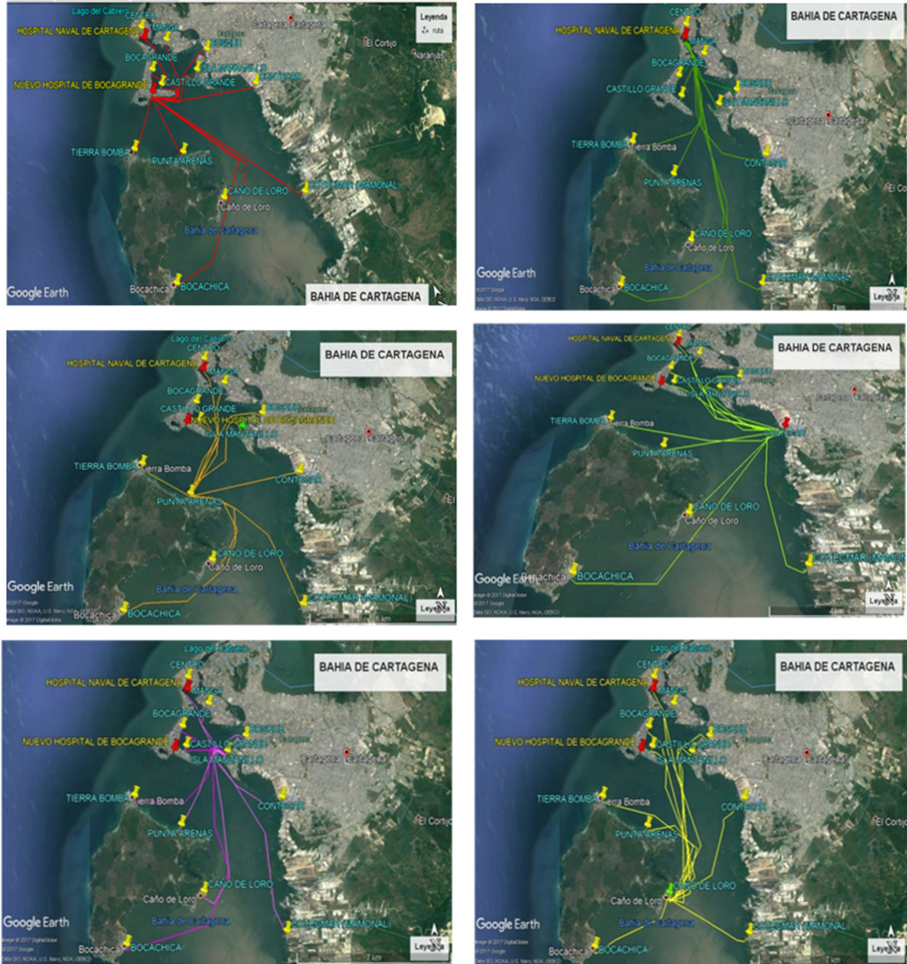


Fig. 1. Map of sea routes of the vulnerable populations in the inner Cartagena Bay

Based on the route distances and the maximum speed of travel (30 knots), a FROM/TO travel time matrix was developed, as displayed in Table 6.

Table 6. FROM/TO matrix of travel time (in minutes) by route between all the nodes. Note: 1) Nuevo Hospital Bocagrande, 2) Hospital Naval, 3) Isla Manzanillo 4) Punta Arenas, 5) Contecar -Mamonal 6) Cotecmar - Mamonal, 7) Caño de loro, 8) Bocachica 9) Tierra Bomba, 10) Castillo grande, 11) Bosque, 12) Manga, 13) Centro and 14) Bocagrande.

From/To	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	0	6,04	3,46	3,42	6,32	9,9	7,88	12,3	2,98	2,6	4,6	4,6	6	3,6
2	5,56	0	3,64	6,54	8	12,1	10,1	16,2	7,58	2,9	3,9	1,54	0,62	2,24
3	3,6	8,04	0	5,34	5,28	10	8,42	13,4	5,84	2,2	1,7	2,54	4,2	2,6
4	3,38	6,8	4,7	0	5,32	8,06	5,68	11	2,76	4,4	5,9	5,76	7,6	5
5	6,04	8,26	5,14	5,38	0	6,88	6,42	11,1	7,66	5,5	6,5	7,12	8,36	6,26
6	9,74	12,6	9,72	7,58	6,84	0	4,6	8,14	11	10	12	11,42	13,4	11,3
7	8,68	12,6	8,28	5,38	5,74	4,76	0	4,7	8,26	8,3	9,4	9,24	11,1	8,54
8	12,8	17	13,4	10	11,1	8,12	4,72	0	13,1	13	14	13,77	15,5	13,1
9	2,98	3,6	6,58	2,66	7,84	11,5	8,48	12,6	0	5,6	6,8	6,64	8,42	5,78
10	2,78	3,04	2,24	4,3	5,76	10,2	7,94	12,9	6,22	0	3,4	2,3	3,56	0,92
11	4,48	4,22	1,72	6	6,64	11,2	9,14	14,1	7,16	3,3	0	2,9	4,82	3,44
12	4,6	1,72	2,74	5,8	7	12	9,7	14,2	6,98	2,3	3	0	2,32	1,78
13	5,78	0,64	3,7	7,18	8,22	13,2	11,1	15,6	8,3	3,5	4,8	2,14	0	2,82
14	3,72	2,36	2,62	4,98	6,62	11,3	8,92	13,9	5,9	0,9	3,5	1,74	2,9	0

The model was programmed in MATHPROG and solved with solver LPSOLVE [6].

When two sea ambulances are available, the application of the mixed integer linear programming algorithms yields an optimal solution:

- Ambulance boat No. 1 should be located in the town of Cotecmar – Mamonal, at coordinates (10 19' 34.79" N, 75 30' 36.05" W).
- Ambulance boat No. 2 should be located in the town of Castillo Grande, at coordinates (10 23' 43.62" N, 75 33' 02.61" W).

As shown in Fig. 2, these ambulance boat locations minimize the response times to emergencies that may arise in the vulnerable populations in the insular area and the Cartagena Bay.

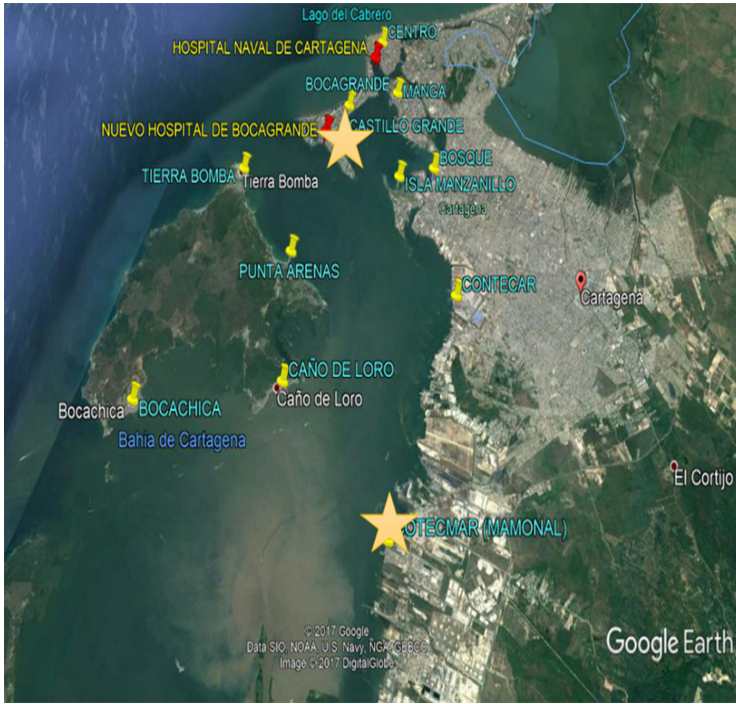


Fig. 2. Optimal location for two ambulance boats in Cartagena Bay

4 Conclusions

An optimal location for ambulance boats was obtained to ensure coverage for vulnerable populations in case of emergencies in the insular area of Cartagena Bay using an MIP model.

The boat location was performed as a function of predefined routes that follow safety standards established by DIMAR, and takes into account a maximum speed of 30 knots.

Two locations were defined: the first one is located on one of the vulnerable populations (towns) in the insular area and the second one is located at a hospital with docks.

These locations provide coverage to any point in the insular area, achieving improvements in response times.

Further research opportunities include expanding the MIP model considering terrestrial ambulances and helicopters, as an interconnected emergency system that optimizes displacements in cities with maritime routes. Additionally, simulation and dynamic location approaches should be considered, including stochastic demand scenarios.

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